

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Dress-o-Tron: Robot-Assisted Dressing in the Real World
<i>Company:</i>	Please provide company name

Contact Details

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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☒ If Applicable: Robot (Hello Robot Stretch), Manikin, Garments, Sensors, GPU workstation
<i>Relevant research areas (if applicable)¹:</i>	AI, Robotics, Computer Vision, Human-Robot Interaction

Project Background Information

A report from the World Intellectual Property Organization estimates that over one billion people worldwide require assistance with daily activities; a figure expected to rise significantly as the global population ages. Caregiving robots offer promising solutions to alleviate the burden on human caregivers while enhancing the independence and quality of life for care recipients. The Dress-o-Tron project aims to advance robotic support for individuals with mobility limitations by focusing on activities of daily living (ADLs), with particular emphasis on dressing, a critical yet often overlooked ADL. Assistive dressing is performed daily by carers for people with limited upper-body mobility and is central to comfort, independence and dignity. It is also technically demanding: the robot must manipulate a highly deformable garment while in sustained physical contact with a person. By developing a prototype robot-assisted dressing system using a mobile manipulator and a seated manikin, first in simulation and then on real hardware, this initiative seeks to explore key challenges in physical caregiving. These include the manipulation of both rigid and deformable objects, and safe, effective physical human-robot interaction. The goal is to contribute to the development of assistive technologies that are both practical and empathetic

in real-world care settings. This project is aligned with the [PhyRC 2027 Challenge](#) (Physical Robotic Caregiving Challenge), jointly organised by the EmPRISE Lab at Cornell University and the Robust Intelligence & Robotics Lab at KAIST. Its single track for 2027 is Mobile Manipulation for Robot-assisted Dressing, evaluated in two phases: a simulation phase (15 September – 15 December 2026) and a real-robot phase (January – May 2027).

Requirements

This project is designed for a resourceful and robotics-passionate team eager to tackle real-world challenges in assistive technology. Successful completion of this project will require a wide range of skills including robotics, AI, image processing and software engineering.

The core task involves developing a robot-assisted dressing system. In this scenario, a manikin is seated in a chair and the robot must dress it in a closed-topology T-shirt: picking up the garment, threading one wrist into a sleeve, pulling the shirt across to the other side, dressing the second sleeve, and maximising the proportion of each arm that ends up covered. The robot must apply gentle, controlled force within safe thresholds to ensure comfort and safety, and must complete the task efficiently, as the challenge rubric divides the score by the time taken and randomises object positions and the manikin's configuration between trials.

Based on the [Stretch robot](#), the simulation environments released by the challenge organisers, students will build a fully integrated system with 1) a navigation module to position the robot accurately relative to the seated care recipient, 2) a perception module to detect and track the manikin's pose and the state of the garment in real time, 3) a manipulation module to grasp a highly deformable garment and route it along the arms with appropriate force and precision, and 4) an orchestration module to coordinate the entire process efficiently and with minimal time. The project involves substantial hands-on work with real hardware and robotic systems, and is expected to conclude with a real-world demonstration of the developed system.

The work splits naturally into two stages that mirror the challenge phases. In the first, the team develops and evaluates a dressing policy in simulation against the published rubric, which awards points for garment pick-up, first sleeve, cross-body pull, second sleeve, and overall arm coverage. In the second, the team transfers the approach to the physical Stretch platform and a manikin in the lab, confronting the sim-to-real gap in perception, contact sensing and cloth dynamics.

Skills

Highly desirable:
Enthusiasm
Interest in healthcare and robotics

Desirable:
Machine Learning
 Programming (Python)
 Robot simulation (NVIDIA Isaac Sim, ROS)

Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.